

Skills Summary

Finding Order and Scale Mistakes on Graphs

Use this reference while completing your worksheet or when studying.

Skill 1 — The order inside an ordered pair

Rule: (x, y) means **across first, then up**. Always.

A reversed pair is not the same place written differently — it is a different place. $(6, 1)$ and $(1, 6)$ are built from the same digits and sit nowhere near each other.

Two reversed pairs lie the same distance either side of the diagonal line where across equals up.

Example: A point should be at $(6, 1)$ but was plotted at $(1, 6)$. Measured left to right, how far apart are the two marks?

Step 1. Left-to-right distance depends only on the across numbers, so I compare the first number of each pair and ignore the up numbers.

Step 2. The across numbers are 6 and 1, so the gap is $6 - 1$. Distance across: **5** units

Try it: A point should be at $(9, 2)$ but was plotted at $(2, 9)$. Measured left to right, how far apart are the marks?

check: 7 units

Skill 2 — What one mark on an axis is worth

Rule: Read the interval before you read the graph.

value = number of marks $\times$ value of one interval

Find the interval by subtracting two labelled marks next to each other. Counting marks and reporting that count is the single most common graph-reading error, and it is always wrong by exactly the interval.

Example: A vertical axis is labelled 0, 4, 8, 12, ... A bar reaches the 6th mark above zero. What value does it show?

Step 1. Two neighbouring labels differ by 4, so one mark is worth 4 — I multiply the mark count by that instead of reading the count itself.

Step 2. $6 \times 4 = 24$, which the labels confirm: the 6th mark is labelled 24. Value shown: **24**

Try it: An axis is labelled 0, 25, 50, 75, ... A bar reaches the 3rd mark above zero. What value does it show?

check: 75

Skill 3 — Totals and means when the axis is scaled

Rule: Convert marks to values *first*, then total or average.

A total built from mark counts is wrong by a factor of the interval — and so is a mean built the same way. Multiplying at the end works too, because every bar shares the same interval, but converting first keeps the numbers meaningful.

Changing the interval changes the picture, never the data.

Example: On a graph whose axis is marked every 10, four bars reach the 2nd, 5th, 3rd and 6th marks. What is the total?

Step 1. Mark counts are not values, so I add the counts and then convert the whole thing with the interval.

Step 2. $2 + 5 + 3 + 6 = 16$ marks in all, and each mark is worth 10, so the total is 16×10 .
Total: **160**

Try it: On a graph whose axis is marked every 5, three bars reach the 4th, 1st and 6th marks. What is the total?

check: 55

(!) Watch out: A taller bar does not always mean a bigger number. Before comparing two graphs, check that both axes use the same interval — redrawing the same data on a finer scale makes every bar taller and changes nothing at all.